

youkined from <https://github.com/Nicicalu/OST-CN2-Spik>

OSPF (IGP)
Equal Cost MultiPath (ECMP) load balancing, fast convergence, widely used. Uses IP Port 89 on Layer 3. Uses 224.0.0.5 (all routers), 224.0.0.6 (all DR,BDR)
Dual Election: Highest interface priority → if tie, highest Router ID. Priority 0 = ineligible for DR/BDR. No preemption. BDR = same rules, second best candidate
ROUTER OPERATION
– Establish neighbor adjacencies and exchange LSAs
– Build the Link-State Database (LSDB)
– Run Dijkstra's SPF algorithm:
– Intra-area change: SPF recalculation
– Inter-area change: No SPF needed → ABR handles updates
– Build the routing table from SPF results
AREAS
– AS split into areas (sub-domains), each with a 32-bit Area ID (e.g., 0.0.0.0 = Area 0)

Backbone Area (Area 0): Core of the OSPF domain (must exist). Must connect to all other areas (directly or via virtual links). Must be contiguous (no disjointed segments). Should not contain end-user networks

Non-Backbone Areas: Connect end users and local resources. All inter-area traffic must transit the backbone

ROUTER TYPES
Area Border Router (ABR): Connects two or more OSPF areas; must have interface in backbone. 1 OSPF DB per Area
Internal Router: All interfaces belong to the same area (non-backbone)
Backbone Router: At least one interface in Area 0. Includes ABRs and routers internal to the backbone
AS Boundary Router (ASBR): Connects to external AS/Network (e.g., BGP). Advertises external routes into OSPF. Can be in backbone or non-backbone area

DESIGN RULES
Rule 1: Backbone (Area 0) must be contiguous → no partitions allowed
Rule 2: Every non-backbone area must connect to Area 0
LSA TYPE OVERVIEW
Type 1 – Router-LSA: Sent by all routers, lists directly connected links (so all outgoing interfaces with state and cost) (intra-area only)
Type 2 – Network-LSA: Sent by DR, lists all routers and DR in broadcast/multi-access network (intra-area only)
Type 3 – Summary-LSA: Sent by ABRs, advertises networks from other areas, flooded in all the areas that are not "totally stubby" (inter-area)
Type 4 – ASBR-Summary: Sent by ABRs, advertises path to ASBRs (inter-area)
Type 5 – AS-External-LSA: Advertises external routes (e.g., BGP); flooded to all non-stub areas → only normal areas
Type 7 – NSSA External: Like Type 5 but used inside NSSA; converted to Type 5 by ABR

AREA TYPES (ALLOWED LSA TYPES)	
Standard (1,2,3,4,5): Normal	Not-So-Stubby Area - NSSA (1,2,3,7):
Stub Area (1,2,3):	(1,2,3,7):
- Blocks external LSAs (Type 5)	- Like stub area, but allows one ASBR inside
- ABR injects a default route (0.0.0.0)	- External routes use LSA Type 7 (converted to Type 5 by ABR)
- Supports LSA Types 1-3	
Totally Stubby Area (1,2):	Totally NSSA (1,2,7):
- Blocks external (Type 5) and summary (Type 3/4)	- Like NSSA but blocks Type 3 so gets default route from ABR
- Only allows default route from ABR	

PACKET TYPES
Type 1 – Hello: Used to discover, maintain, and verify neighbors; forms adjacencies. Also for election of DR,BDR in broadcast networks. Contains network mask(of sending routers' interface), Hello interval(p2p,broadcast: default=10s), Options, Priority(for election), Router dead interval(default=40s), DR/BDR IP, Neighbors
Type 2 – Database Description (DBD/DD): Exchange summaries of LSAs during adjacency formation (headers only). Contains Interface Max. MTU, Options, I/M/MS bits (Initial, More, Master-slave bit), DD Sequence num, LSA Header
Type 3 – Link State Request (LSR): Sent when a router needs specific LSAs listed in the DBD. Contains Link State Type (router/network), Link State ID, Advertising Router (sender address)
Type 4 – Link State Update (LSU): Used to flood new or updated LSAs. Contains Number of LSAs, full LSAs information
Type 5 – Link State Acknowledgment (LSAck): Confirms receipt of LSAs to ensure reliable flooding. For this you send LSAck or implicitly by sending LSU with same info back. Many acks may be grouped together to a single LSAck.

SUB-PROTOCOLS
Hello Protocol:
– Used for neighbor discovery and parameter negotiation.
– Maintains logical adjacencies on P2P, P2MP, and virtual links.
– Elects DR/BDR on broadcast and NBMA networks.
– Continuously sends hello packets to maintain bidirectional connectivity; failure to receive = neighbor down (in agreed router dead interval at initialization)
Database Sync Protocol:
– Syncs LSDB using Database Description (DBD) packets with only LSA headers.
– Uses I-bit (initial), M-bit (more), and MS-bit (master/slave).
– ExStart: Bi-direction; highest Router-ID = master. Determine initial seq nr
– Exchange: Exchange of DBD packets (LSA headers).
– Loading: Missing LSAs are requested.
– Full: Databases fully synchronized.

OSPF ROUTING AND ECMP
– Each router runs Dijkstra per area; link cost = metric from LSAs (1–65535)
– OSPF prefers more specific match (CIDR) and if then still multiple: intra-area > inter-area > external
– Routes added to RIB/FIB based on computed next hops
– ECMP: Modified Dijkstra supports Equal-Cost MultiPath if multiple paths have same cost → routes added with multiple next-hops for load balancing

OSPF ROUTE SELECTION
Intra-Area (O): Source and dest in same area; routes from Type 1 and 2 LSAs
Inter-Area (O IA): Source and dest in different areas within same AS; via Type 3 LSAs through backbone

External (E1/E2): Dest outside AS; info injected by ASBR via redistribution
– E1: Total = external + internal OSPF cost
– E2: Only external cost (default)
Preference order: More specific route > Intra-area > Inter-area > E1 > E2
Cost calculation: Cost = Reference Bandwidth/default 100 Mbps/Interface Bandwidth
IS-IS (IGP)
CLNS – CONNECTIONLESS NETWORK SERVICES
– CLNS: ISO Layer 3 datagram service; supports CLNP ES-IS, IS-IS.
– CLNP: Connectionless Network Protocol, similar to IP; used in ISO stack (EtherType 0xFFE).
– IS-IS: Link-state routing protocol (Layer 3); forms adjacencies with ES-IS; designed for CLNP but extended (Integrated IS-IS) to support IP.
– Integrated IS-IS: Allows IP routing with IS-IS; used widely by service providers even without CLNP.

NET ADDRESSING (TYPE OF NSAP ADDRESS)
NSAP: Network Service Access Point
NET = Network Entity Title: Unique router identifier in IS-IS
Format: 49.AAAA.BBBB.BBBB.0000.00.AAAA... Area ID (variable length)
BBBB.BBBB.BBBB: System ID (usually 6 bytes = unique router ID)
– 0: N-selector (NSEL) (always 00 for routers)
System ID: Must be unique per router (often based on lo-IP/MAC)
– Example: 49.0001.1921.6800.1024.00 based on IP 192.168.1.24
Area ID: Used for routing hierarchy (like OSPF areas)
IS-IS PACKET TYPES (ALL IN L1 OR L2)
IIH (Hello): Builds and maintains adjacencies; includes system ID, holding time, prio
– Built from 3 functions: discover, build, maintain
– Interval: 10s (default); DIS sends every 3.3s on LANs
– Multipier: Missed Hello limit → Holdtime + interval * Multiplier(default=3)
Multicast:
– 01-80-c2-00-00-14 (All L1ISs)
– 01-80-c2-00-00-15 (All L2ISs)
LSP (Link State PDU): Contains topology info including prefixes with costs; flooded throughout the area → similar to OSPF LSA Type 1
CSNP (Complete SNPP): Sent by DIS; lists all known LSPPs (used for database sync)

PSNP (Partial SNPP): Used to request missing LSPs or acknowledge received LSPs
IS-IS Packet Structure: Common header + TLVs
ROUTER TYPES
L1 Router: Only within one area; no inter-area routing
L2 Router: Backbone route; routes between areas
L1/L2 Router: Acts as both; separates databases, redistributes between levels
BROADCAST/MULTI-ACCESS LINKS: DIS – DESIGNATED IS
– Required on broadcast links (no DIS on p2p)
– Sends periodic CSNPs to ensure DB sync, creates pseudonode LSP
– No backup DIS in IS-IS

DIS ELECTION
– 1) Highest interface priority (0–127) Cisco default = 64
– 2) Highest SNPA (MAC-Address)
Preamption: Enabled – higher prio router automatically takes ver the DIS Role
POINT-TO-POINT LINKS (NO DIS)
CSNP: Sent once at adjacency startup
LSP: Advertises topology changes (link-state info)
PSNP: Acknowledges received LSPs or requests missing ones
PATH SELECTION
Path Selection Order (in IS-IS) – Lower Metric better:
– L1 intra-area routes
– L2 intra-area routes
– Leaked L2 → L1 (internal metric)
– L1 external (external metric)
– L2 external (external metric)
– Leaked L2 → L1 (external metric)
LEVEL 1 ROUTING
– Intra-area routing only (like OSPF intra-area)
– L1 routers use the closest L1/L2 router for inter-area traffic
L1/L2 routers:
– Do not advertise L2 routes into the L1 area (unless route leaking is active)
– Set Attached bit to signal L2 connectivity to backbone
– L1 routers install a default route to nearest L1/L2
– L1 area like OSPF Totally Stubby Area
– Distribution Bit: Set to 'up' (1) on L2 → L1 leaks; blocks re-advertisement L1 → L2.
– Route-Leaking injects a more specific route into L1 to improve routing

LEVEL 2 ROUTING		
– Routing between areas (inter-area)		
– L1/L2 routers inject L1 routes into L2 topology		
– L1 routes are redistributed into L2 with L1 metric preserved in L2 LSP		
IS-IS VS OSPF		
Feature	IS-IS	OSPF
Layer	L2 (CLNS)	L3 (IP, proto 89)
Encapsulation	No IP, uses TLVs	IP packets
Hello Type	IIH	Hello packet
Area Model	L1/L2	Backbone + Areas
Metric	Cost (default 10)	Cost (bandwidth)
Router ID	System ID (6B)	32-bit Router ID
Adj. Types	L1, L2, L1/L2	DR/BDR, P2P
LSDB	Per level (L1/L2)	Per area
Scaling	Large-scale ISP core	Enterprise/campus
Routing info	TLVs (flexible)	Fixed LSA types
BGP (EGP)		
CONFIG		
– next-hop-self fixing iBGP: neighbor (neighbor IP) next-hop-self (when overriding)		
BGP SESSIONS		
Point-to-point adjacencies between BGP routers		
– iBGP: Between routers in the same AS, AD=200, more trusted (lower security overhead)		
– eBGP: Between routers in different ASes, AD=20, stricter policy enforcement		
AUTONOMOUS SYSTEM NUMBERS (ASN)		
– Unique ID for each AS; required for Internet routing with BGP		

– Private ranges:
– 64512–65535 (legacy 16-bit)
– 4200000000–4294967294 (32-bit)
BGP PEERING / NEIGHBORS
– Two routers with a BGP TCP session (port 179) are called peers or neighbors
– Each BGP router is a BGP speaker
– BGP exchanges routing info between ASes (loop-free, policy-based)
– Supports CIDR, route aggregation; decisions based on policies/rules
PATH ATTRIBUTES
– Used for route control and policy enforcement
Well-known mandatory: Always present (e.g., AS-Path, Origin, Next Hop)
Well-known discretionary: Optional but recognized by all (e.g., Local Pref, Atomic Aggregate)
Optional transitive: Passed between ASes (e.g., Community, Aggregator)
Optional non-transitive: Not passed across ASes (e.g., MED, Weight, Originator ID, Cluster ID/list)
NLRI: Routing table info: prefix, prefix length, and associated path attributes
LOOP PREVENTION
– BGP uses AS-Path (list of ASNs) to detect loops. If a router sees its own ASN in a received route, it discards it.
AS-Override: Allows reuse of same ASN across different customer sites (e.g., Swisscom); rewrites ASN to avoid loop detection

BGP MESSAGES
OPEN: Establishes session; includes version, ASN, Hold Time, BGP Identifier, optional params
– Hold Time: Heartbeat in seconds (default 180s, Cisco), reset by KEEPALIVE/UPDATE; 0 = session down
– BGP Identifier: 32-bit Router-ID, manually set or highest loopback/active IP; used for loop prevention
KEEPALIVE: Sent every 1/3 of Hold Time (default 180s); ensures neighbor liveness (BGP doesn't rely on TCP keepalive)
UPDATE: Advertises new routes, withdraws old ones, or both; includes NLRI (prefix + path attributes); can act as KEEPALIVE
NOTIFICATION: Sent on session error (e.g. hold timer expired); terminates session immediately
BGP NETWORK STATEMENTS
– Purpose: Advertise specific prefixes to BGP peers (does not activate interfaces)
– Prefix must exist exactly in the RIB (from static, connected, or learned route)
– Attributes (e.g., origin, next-hop, MED) depend on how the route exists in RIB
– BGP advertises only the best path for a prefix to peers, even if multiple exist
BEST PATH CALCULATION
– BGP maintains all received paths per prefix but advertises only the best one
– Best path is installed in RIB; recalculated on:
– Next-hop reachability change
– Interface failure to eBGP peer
– Redistribution change
– New/withdrawn path received
– Influence:
– Outbound BGP policy → inbound traffic behavior
– Inbound BGP policy → outbound traffic behavior

BGP BEST PATH SELECTION (IN ORDER):
1) Prefer highest Weight (Cisco-specific, local to router)
2) Prefer highest Local Preference (global within AS)
3) Prefer routes originated by the router (only small i in path, NH 0.0.0.0)
4) Prefer shorter AS path (only length is compared)
5) Prefer lowest origin type: IGP < EGP < Incomplete (I on origin)
6) Prefer lowest MED (Multi-Exit Discriminator) (also called metric)
7) Prefer external (eBGP) over internal (iBGP)
8) For iBGP: prefer path with lowest IGP metric to next-hop
9) For eBGP: Prefer oldest (more stable) path
10) Prefer lowest BGP router ID
11) Prefer path from lowest neighbor IP address
ROUTE FILTERING
– Filters control which routes are received/advertised
– Used for security, traffic shaping, memory optimization
– Tools: prefix-list (IP), filter-list (AS-path), route-map (flexible match/set)
BGP COMMUNITIES
– 32-bit optional, transitive tag (e.g. ASN-value, 65000:100)
– Used to mark routes for policy control across ASes
– Can be added, modified, or removed at each hop
iBGP SCALABILITY
– iBGP does not re-advertise routes between iBGP peers → full mesh required (cause no loop prevention)
– Session count = $\frac{n(n-1)}{2}$ (e.g., 5 routers = 10 sessions, 10 = 45)
ROUTE REFLECTORS (RR)
– Solves iBGP full-mesh scaling by allowing selective route reflection
– Clients only peer with RR; unaware they're clients
– RR Rules:
– From non-client → advertise to clients only
– From client → advertise to all (clients & non-clients)
– From eBGP peer → advertise to all (clients & non-clients)
– Only the RR needs special config – clients remain unaware of route reflection. This eliminates the need for full iBGP mesh.

PEERING VS. TRANSIT
Transit: ISP provides full reachability (paid relationship)
Peering: ISPs exchange selected routes; equal relationship, usually unpaid
AS Path Filtering: to avoid getting transit; ip as-path access-list 10 permit ^ \$
INTERNET EXCHANGE POINT (IXP)
– Facility where networks exchange traffic via BGP peering
– Reduces transit costs, latency, and offloads upstream links
PUBLIC PEERING
– Members peer via shared switch fabric and a route server
– Route server distributes routes but stays out of data path (NEXT_HOP unchanged)
– Minimal policy control; one BGP session to route server
– Simplified setup (one legal contract)
PRIVATE PEERING
– Direct BGP sessions between two parties (1:1)
– May use public or private interconnects
– Full policy control per neighbor
– Requires one session and legal contract per peer
Examples: Equinix, SwissIX (non-profit)
ENTERPRISE CONNECTIVITY OPTIONS
Single-Homed: One ISP, one link (BGP or static); simple but no redundancy

Dual-Homed: One ISP, two links (or routers); redundancy within same provider
Multihomed: Multiple ISPs; improved redundancy and routing control, but avoid being a transit – advertise only customer-owned prefixes
Dual-Multihomed: Multihomed, but two links per ISP
TRAFFIC ENGINEERING (TE)
Outbound TE (Local Pref): Set higher local pref to prefer exit path; affects outbound traffic (highest wins)
Inbound TE (MED): Signal entry preference with MED; lowest wins; only works if peer honors it
Inbound TE (AS-Path Prepending): Add own ASN multiple times on backup path; shortest AS-path wins
TE Limitation: Affects only routes outbound (e.g. local pref); inbound control limited – ISPs may ignore MED
TE with Aggregate: Prefer primary ISP with summarized routes; advertise specific prefixes on backup for fallover
Aggregate Impact: Longest-match wins → specific prefixes may steer traffic to alternate ISP; avoid provider-owned aggregates
RPKI – RESOURCE PUBLIC KEY INFRASTRUCTURE
– Framework to validate that a prefix is legitimately originated by a specific ASN
– Prevents route hijacking and accidental mis-originations (route leaks)
– RPKI validates origin (not just the 5 RRs)
KEY COMPONENTS
Trust Anchors (TA): Root CAs of the AS-Paths; issue certs for resource holders
ROA – Route Origin Authorization: Digitally signed object that authorizes ASN to announce a prefix (contains AS, prefix that AS can originate, max prefix length)
RPKI Validators: Software that downloads, verifies, and stores Validated ROA Payloads (VRPs)
RPKI VALIDATION STATES
Valid: Prefix + ASN match ROA
Invalid: Prefix found, but ASN mismatch or prefix too long
Not Found / Unknown: No matching ROA
VALIDATOR DETAILS
– Use TALS (Trust Anchor Locators) to fetch data from RIRs
– Validate cryptographic signatures (via X.509 certs with RFC 3779)
– Outputs VRPs; invalid objects are discarded
– Update frequency: ~24h (recommended), ~30–60min (practice)
– RRDP (RFC 8182) replacing rsync (uses HTTPS)
BGP MONITORING
– Event tracking, BGP hijack detection, Route leak detection, RPKI status check, Reachability tracking, AS path change tracking, AS path visualization

NETWORK DESIGN
Pillars: Scalability, Speed, Availability, Security, Manageability → overall Cost
AVAILABILITY CONCEPTS (MOST IMPORTANT REQUIREMENT)
MTBF: Mean Time Between Failures; MTTR: Mean Time to Repair
$- A = \frac{MTBF}{MTBF + MTTR}$ MTBF combined: $\left(\sum_{n=1}^N \frac{1}{MTBF_n} \right)^{-1}$ parallel; $\frac{MTBF_n}{n}$
– Lower MTTR + higher MTBF = more availability
REDUNDANCY
– Adds reliability, decreases MTBF but increases MTTR and complexity
– Balance: resilience vs. manageability
Backup Paths: Duplicate devices/links on primary path, build extra links for redundancy, consider backup link capacity, consider failover speed
Load Balancing: ECMP, EtherChannel, Port-Channel
HIERARCHICAL DESIGN
Access: Connect end devices, high port count, port security, L2, QoS marking
Distribution: L3, policy control, HSRP/VRP, loop protection, small fault domain
Core: High-speed backbone, L3 only, no policy, scalable/redundant, no security
Collapsed Core: Combines Core + Distribution (small/medium networks)
FLAT DESIGN
– Any-to-any; small networks, MPLS/LAN setups
– Lacks scalability and control
FABRIC DESIGN
– Modern design using Underlay/Overlay
Underlay: transport (e.g., IP, MPLS)
Overlay: logical virtual topology (e.g., EVPN)
ENTERPRISE CAMPUS
– 100s/1000s of users, multiple buildings, one physical location
– Multiple interconnected LANs, connected via Ethernet and Wireless
FHRP – FIRST-HOP REDUNDANCY PROTOCOLS
– Ensures default gateway is always reachable, PCs can only have one
– Enables fast failover during router failure

VRPP – Virtual Router Redundancy Protocol (Multivendor): Shared virtual IP, real MACs per router, Master router handles forwarding
HSRP – Hot Standby Router Protocol (Cisco): Shared virtual IP + virtual MAC, One active, others in standby → faster
GLBP – Gateway Load Balancing Protocol (Cisco): Load balancing + redundancy, Shared virtual IP + multiple virtual MACs, Roles: AVG: Answers ARP and sends virtual MAC addresses of AVFs AVF: Forwards traffic
DATA CENTER DESIGNS
ToR – Top of Rack: switches per rack, less cabling, easy expansions/ exchanges per 'rack', scalable glass fiber, ideal for high service density (full racks). But more switches, more ports, more L2 Srv-2 Srv traffic, more STP to be managed
EoR – End of Row: 1 switch per row, less switches, higher utilization of ports, switches all at one place, better L2 availability between racks. But more cabling
TRAFFIC DIRECTIONS
North-South: Between external networks (client-server, in/out DC)
East-West: Within DC (e.g., server-server, storage)
THREE-TIER DC
– Access – Aggregation – Core (like Hierarchical)
– Optimized for North–South traffic
– Not ideal for East–West communication
LEAF-SPINE ARCHITECTURE
– Two-tier: Leaf switches (access) connect to Spines (core)
– High performance, low latency
– Scalable, ideal for East–West traffic

MULTICAST
Use-cases 1-to-Many: Streaming, software updates, music-on-hold
Use-cases Many-to-Many: Gaming, VR, stock data, group chat
Benefits: Efficient bandwidth, lower server/CPU load, no redundancy, supports multipoit apps
Properties: UDP-based (no delivery guarantee, congestion control, or ordering) Apps must handle drops, duplicates, out-of-order packets
Source: Sends to group IP; doesn't need to join
Receiver: Must explicitly join group to receive traffic
MULTICAST ADDRESS RANGES
224.0.0.0 – 224.0.0.255: Link-local, TTL = 1 (not forwarded by routers)
224.0.1.0 – 224.0.1.255: Reserved by IANA, routable
224.0.0.0 – 239.255.255.255: Source-Specific Multicast (SSM)
239.0.0.0 – 239.255.255.255: Administratively scoped (private multicast space)
BROADCAST BASICS
L3 routes between subnets; L2 floods within same subnet
L2 (Bridging): MAC flit.filt.filt; switches flood to all ports in VLAN
L3 (Routing):
– 255.255.255.255: local broadcast, not routed
– Directed broadcast (e.g. 10.1.1.1.255): can be routed if enabled
L2 MULTICAST: MAC MAPPING
1) Get IP: Example multicast IP address: 239.5.5.5
2) Convert to Binary: 239.5.5.5 = 11011111.00000101.00000101.00000101
– Take only last 23 bits: 00000101.00000101.00000101
3) Map to MAC Prefix: Use fixed MAC multicast prefix: 01:00:5E → Map last 23 bits to: 01:00:5E:05:05:05
4) Final MAC Address: 01:00:5E:05:05:05
IGMP – INTERNET GROUP MANAGEMENT PROTOCOL (HOST TO FIRST-HOP-ROUTER)
Purpose: Manages group membership for IPv4 multicast on each segment

IGMPv1:
– Basic join via query-response mechanism
– No way for a host to leave a group explicitly
– Router sends general membership queries every 60s to 224.0.0.1
– If no report is received, router removes group after timeout
– Receiver has no knowledge of the multicast source
IGMPv2:
– Adds Leave Group message (faster pruning of unused traffic)
– General queries to 224.0.0.1 every 125s 'Any hosts interested in any groups?'
– Supports Group-Specific Queries e.g. when someone leaves group ('anyone still interested in group x?'), reducing broadcast overhead
– Still source-agnostic: receivers don't know who the source is
IGMPv3:
– Adds source filtering (Include/Exclude lists)
– Enables Source-Specific Multicast (SSM) – receiver requests traffic only from selected source(s), no need for Rendezvous Point (RP) anymore
– Adds support for applying source-level access control (source filtering)
– Can also be used in ASM (Any Source Multicast), but mainly with SSM

IGMP SNOOPING
Without snooping: multicast = broadcast on VLAN
With snooping: switch listens to IGMP messages and builds a forwarding table
Default: snooping is enabled; switch needs IGMP Query to operate
REVERSE PATH FORWARDING (RPF)
RPF Check: To avoid loops, verifies that a multicast packet arrives on the interface that a unicast packet destined for the multicast source would be forwarded out of.
RENDEZVOUS POINT (RP)
– Used in Shared Tree (*,G) setups with PIM-SM
– RP acts as the common meeting point for sources and receivers
– RPF check is performed toward the RP (not the source)
– Once the source is known, routers may switch to a Source Tree (S,G)
SHARED TREE VS. SOURCE-BASED TREE
Shared Tree (*,G):
– IGMP host sends a membership report (IGMP Join)
– Router adds (*,G) entry to multicast routing table
– * means 'any source' → source is unknown/unspecified
Source-Based Tree (S,G):
– Built when router receives an (S,G) join/report from IGMP host
– S = known multicast source; G = group
– Router adds (S,G) to mroute table once source is known

PIM – PROTOCOL INDEPENDENT MULTICAST (ONLY BETWEEN ROUTERS)
– Relies entirely on the unicast routing table (RIB) for multicast forwarding decisions
– Protocol-independent: works with static routes, OSPF, IS-IS, etc.
PIM-DH – DENSE MODE (NO RP)
Push Model: Floods multicast traffic to all interfaces; then prunes where no receivers exist

1) Flooding: Source sends traffic → forwarded out all multicast-enabled links using unicast RIB
2) Distribution Tree: Initially includes entire network (shared tree rooted at source)
3) Prune Messages: Routers without interested receivers send prunes upstream to remove themselves from the tree
4) State Maintenance: Routers track source, receivers, interfaces to forward/prune per group
PIM-SM – SPARSE MODE (ASM, SSM)
Pull Model: Multicast traffic is only sent where requested. Works with IGMP to detect interested receivers and uses unicast routing for forwarding.
1) Join/Purge: Routers send explicit join/prune messages to request or stop receiving multicast for group (G) to other routers
2) Forwarding: Routers only forward multicast packets for group (G) on interfaces from which explicit joins were received
ASM – ANY SOURCE MULTICAST (HOW IT WORKS WITH RP)
– Works with IGMPv1 or IGMPv2 → receiver does not know the source
– Receiver sends IGMP Join (*,G) to its first-hop router
– First-hop router forwards PIM Join (*,G) hop-by-hop toward the Rendezvous Point (RP)
– RP acts as a common meeting point for sources and receivers
– Sources send multicast traffic to the RP via a PIM Register tunnel
– All routers in the multicast domain must know the RP location

HOST DELETION & MOVE

Host Deletion: When a host detaches, its ARP (default: 1500s) and MAC entry (default: 1800s) time out on the VTEP. Upon aging, the VTEP withdraws the host's MAC/L2VNI and IP/L3VNI advertisements.

- imp-null = networks are directly connected, no more label switching

WAN

Connects remote LANs via SPs for data/voice/video; key needs: bandwidth, control, design, resilience, mgmt. **Requirements:**

Bandwidth: App needs, peak usage, reserve for VoIP
Control/Security: Trust provider? No full control
Availability: Redundancy, SLA for failures
Mgmt: Inband vs out-of-band

Global Segments: Known and supported by all SR nodes in the domain, Installed in forwarding tables across the network (e.g. "Forward packet according to shortest path to Node1")

Local Segments: Defined and installed only on originating node, Not for-

Global segments: Defined in the SR Global Block (SRGB) and should be consistent across all nodes; **local segments** are defined in the SR Local Block (SLB) and are specific to the local SR node.

Service Policy: Apply policies to interfaces or directions (in/out)

CDN

Origin Server: Central content source (original files), usually in a datacenter

Edge / CDN Server (POP - Point of Presence): Geographically distributed caches content

DNS Infrastructure: Directs users to optimal edge server (e.g. via Geo-
Routing)

KEY BENEFITS

- Latency Reduction:** Nearby edge servers reduce round-trip time
- Availability:** Failover and redundancy in case of node failure
- Scalability:** Handles traffic spikes via load balancing
- Cost Optimization:** Reduces backend and transit load on origin
- DDoS Protection:** Edge servers absorb attacks – not all traffic on one server
- Global Load Reduction:** Less long-distance traffic across the Internet

EVPN BGP ROUTING TABLE INFOS

* = Would not be there if it was L2 VNI BGP Routing Table

Route Distinguisher: 172.16.255.101:32777

Route Type: 2

MAC Address Length: 48

```
AS-path: NUNE, path sourced internal to AS
172.16.254.101 (metric 81) from 172.16.255.1 (172.16.255.1)
Origin IGP, MED not set, Localpref 100, weight 0
Received label 30010 50000
Extcommunity: RT:1:10 RT:65000:50000 ENCAP:8 Router MAC:5254.00ca.69ae
Originator: 172.16.255.101 Cluster list: 172.16.255.1
```

Campus Design: used to reduce size of L2 domain: EVPN, MPLS

- VXLAN is a data plane technology which encapsulates Ethernet frames in UDP datagrams to tunnel layer 2 frames over a layer 3 network.
- The underlay network is unaware of VXLAN devices that connect to the physical switches are unaware of VXLAN.
- A route distinguisher is used to uniquely identify a route in combination with the destination prefix.